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# Problem Statement Title

- Track:4
- PS ID:402
- Team Name:DATA DYNAMOS
- Team Leader name:JIGISHA PANCHAL

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# Detection of Digital Threats & Malicious Content

- Core Innovation: A multi-layered AI framework that provides sub-second threat detection and autonomous, goal-driven response.
- Unified Defense: Synthesizes intelligence from critical domains: Network, Application and Content.
- Explainable AI (XAI): Converts complex machine learning alerts into human-readable summaries using LLMs for transparent decision-making.
- Agentic Response: Moves beyond monitoring to active defense—automatically blocking IPs and isolating compromised systems.

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# Technical Approach

- Model Training & Detection: Custom training of domain-specific models (LSTM for Network, BERT for NLP) using curated cybersecurity datasets. Implementation of an Inference Engine to process live telemetry and identify multi-vector attacks in real-time.
- Monitoring & Visualization UI: Centralized Security Dashboard developed for real-time threat visualization and log management. Interactive logs showing threat classification, severity levels, and origin.
- Deployment & Infrastructure: GCP (Google Cloud Platform): Hosting the heavy-lift AI models and backend processing for high availability. Vercel: Deploying the frontend for a fast, responsive, and global user interface.

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# Feasibility & Viability

- Technical Feasibility: Leveraging pre-trained models (Transfer Learning) allows for high-accuracy deployment within a 24-hour window.
- Scalability: The microservices-ready design ensures the prototype can easily migrate to enterprise tools like Elasticsearch post-hackathon.
- Potential Challenges: High-volume data ingestion in a short timeframe.
- Risk Mitigation: Focusing on a "Golden Path" demo—showing a specific multi-vector attack (e.g., a phishing email leading to a malicious process) to prove the correlation engine works.

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# Impact & Benefits

- Drastic Latency Reduction: Achieves a Mean Time to Detect (MTTD) of <10 second, stopping threats before they propagate.
- Operational Efficiency: Automates Tier-1 SOC tasks, aiming for a 70% reduction in manual analyst workload.
- Proactive Defense: Shifts the enterprise posture from "Reactive Monitoring" to "Autonomous Prevention," decreasing successful breaches by an estimated 85%.
- Transparency: Uses LLM-generated summaries to provide Explainable AI (XAI), ensuring security teams understand why a threat was blocked.
- Holistic Visibility: Provides a single "Source of Truth" across Network, App, and Human layers, closing the gaps left by siloed legacy tools.

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# References


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